

Journal Scope Drift Report

CWTS Leiden citation clustering · macro level · baseline 2020 → 2026 · run 20260721_122750

38,222 Papers analysed	8.1% Overall out-of-scope	3 / 8 High-drift journals	0.289 Mean JSD 2026
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Summary

This report looks at 8 Frontiers journals and asks a simple question: is the type of research each one publishes drifting away from where it started in 2020? Across all of them, about 1 in 12 papers (8.1%) now falls outside the journal's core subject areas. Three journals — Materials, Neurorobotics and Environmental Science — have moved the furthest from their original focus. A page of plain-language commentary and supporting detail follows for each journal.

Scope Drift & OOS Summary by Journal

Journal	Papers	OOS	OOS %	JSD 2026	Drift
Fr. Neurorobotics	1,037	229	22.1%	0.378	High
Fr. Robotics and AI	1,791	207	11.6%	0.314	Moderate
Fr. Chemistry	6,003	649	10.8%	0.239	Low
Fr. Environmental Science	7,093	678	9.6%	0.343	High
Fr. Materials	3,123	246	7.9%	0.379	High
Fr. Surgery	5,885	407	6.9%	0.232	Low
Fr. Earth Science	7,870	401	5.1%	0.280	Moderate
Fr. Aging Neuroscience	5,420	271	5.0%	0.145	Low

A note on how to read this report: This analysis is based purely on citation patterns (which papers cite which), not on reading the papers themselves. Citation-based methods like this can misclassify certain paper types — methods papers and meta-analyses in particular tend to cite (and be cited by) a much wider range of fields than their own subject matter would suggest, so they can appear more “out-of-scope” than they really are. Scope itself is also defined statistically here: a journal's “primary” clusters are simply whichever citation communities together account for 80% of its papers. This threshold is applied uniformly across all journals and hasn't been tuned per journal, so some journals may need a different cutoff to reflect their actual editorial scope. Treat the figures below as a useful signal for further investigation, not a definitive scope audit.

Frontiers in Materials

High

Materials shows the clearest signs of scope movement in this review. Compared with where the journal stood in 2020, the mix of research it publishes has shifted noticeably — a change of this size (JSD 0.38) means the papers coming in today look meaningfully different from the ones that defined the journal a few years ago.

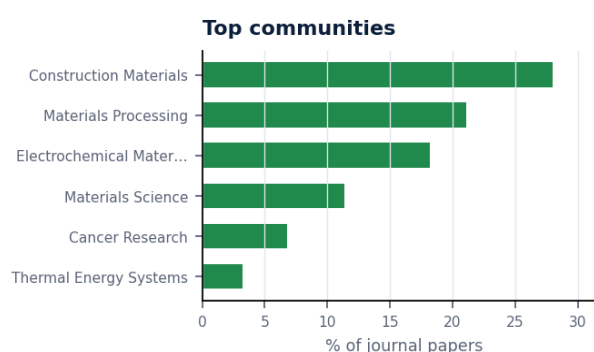
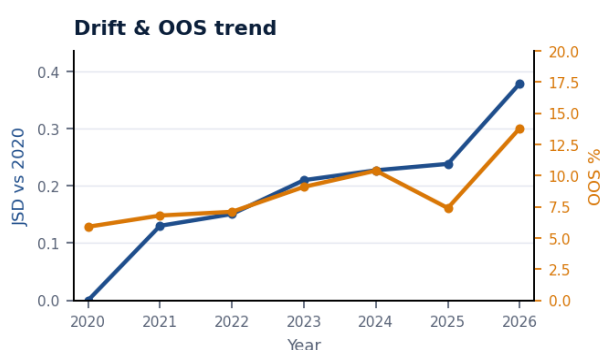
Right now, about 8% of Materials's papers (246 out of 3,123) fall outside its core subject clusters — mostly work related to Dental Health, Organic Synthesis. This out-of-scope share has risen since 2020. At the same time, the journal is concentrating more tightly around a smaller set of themes rather than spreading out. Its largest single topic area today is “Construction Materials.”

Scope split

92% in-scope

8% OOS

3,123 Papers	246 (7.9%) Out-of-scope	0.379 JSD 2026	-0.228 Entropy Δ	5 Primary clusters	80.0% Primary coverage
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Community composition

Community	Status	Papers	Share of journal
Construction Materials	In-scope	875	28.0%
Materials Processing	In-scope	660	21.1%
Electrochemical Materials	In-scope	567	18.2%
Materials Science	In-scope	355	11.4%
Cancer Research	In-scope	213	6.8%
Thermal Energy Systems	In-scope	100	3.2%

Example papers

In-scope

- Influence of surfactant on conductivity, capacitance and doping of electrodeposited polyaniline films
- The 4-Octahedra model
- Study on the Residual Stress Relieving Mechanism of C/C Composite-Nb Brazed Joint by Employing a Structurally Optimized Graphene Reinforced Cu Foam Interlayer
- Green polymer nanocomposites: bridging material innovation with sustainable industrial practices
- Fabrication of Mg-Al Layered Double Hydroxide Thin Membrane for All-Solid-State Alkaline Fuel Cell Using Glass Paper as a Support

Out-of-scope

- Comparison Between Different Composite Resins Used for Clear Aligner Attachments: An In-Vitro Study
- Processing–Structure–Properties of Cork Polymer Composites
- Fungal-derived functional carbons for secondary batteries: from biomass architectures to electrochemical performance
- Experimental Investigation on Uniaxial Unconfined Compressive Properties of Ice on Asphalt Pavement Surface
- Phenacyl bromide as Norrish type I photoinitiator for the facile synthesis of chain-end functional PMMA and polystyrene

Frontiers in Neurorobotics

High

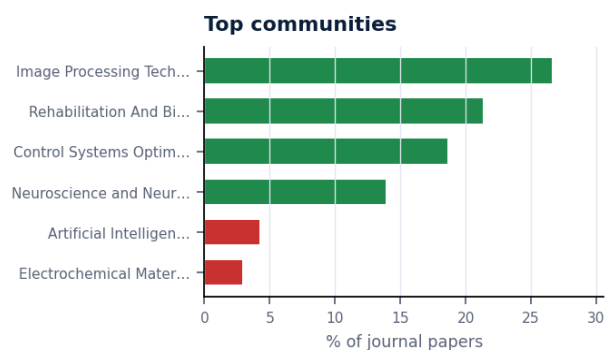
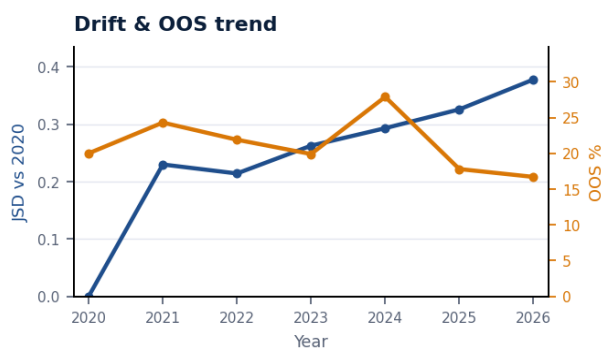
Neurorobotics shows the clearest signs of scope movement in this review. Compared with where the journal stood in 2020, the mix of research it publishes has shifted noticeably — a change of this size (JSD 0.38) means the papers coming in today look meaningfully different from the ones that defined the journal a few years ago.

Right now, about 22% of Neurorobotics's papers (229 out of 1,037) fall outside its core subject clusters — mostly work related to Artificial Intelligence Applications, Electrochemical Materials. This out-of-scope share has fallen since 2020. At the same time, the journal is concentrating more tightly around a smaller set of themes rather than spreading out. Its largest single topic area today is “Image Processing Techniques.”

Scope split

78% in-scope
22% OOS

1,037 Papers	229 (22.1%) Out-of-scope	0.378 JSD 2026	-0.4157 Entropy Δ	4 Primary clusters	80.0% Primary coverage
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Community composition

Community	Status	Papers	Share of journal
Image Processing Techniques	In-scope	276	26.6%
Rehabilitation And Biomechanics	In-scope	221	21.3%
Control Systems Optimization	In-scope	193	18.6%
Neuroscience and Neurodegeneration	In-scope	144	13.9%
Artificial Intelligence Applications	Out-of-scope	44	4.2%
Electrochemical Materials	Out-of-scope	30	2.9%

Example papers

In-scope

- Editorial: Cognitive Brain-Inspired Cyber-Physical Systems in Industrial Informatics
- Integrated Circuit Board Object Detection and Image Augmentation Fusion Model Based on YOLO
- Resolving uncertainty on the fly: modeling adaptive driving behavior as active inference
- Learning Then, Learning Now, and Every Second in Between: Lifelong Learning With a Simulated Humanoid Robot
- Adaptive Locomotion Control of a Hexapod Robot via Bio-Inspired Learning

Out-of-scope

- Gaze-Based Intention Estimation for Shared Autonomy in Pick-and-Place Tasks
- A Clustering Algorithm for Multi-Modal Heterogeneous Big Data With Abnormal Data
- Optimal UAV's Deployment and Transmit Power Design for Two Users Uplink NOMA Systems
- Application of Intelligent Inspection Robot in Coal Mine Industrial Heritage Landscape: Taking Wangshiwa Coal Mine as an Example
- The use of deep learning technology in dance movement generation

Frontiers in Environmental Science

High

Environmental Science shows the clearest signs of scope movement in this review. Compared with where the journal stood in 2020, the mix of research it publishes has shifted noticeably — a change of this size (JSD 0.34) means the papers coming in today look meaningfully different from the ones that defined the journal a few years ago.

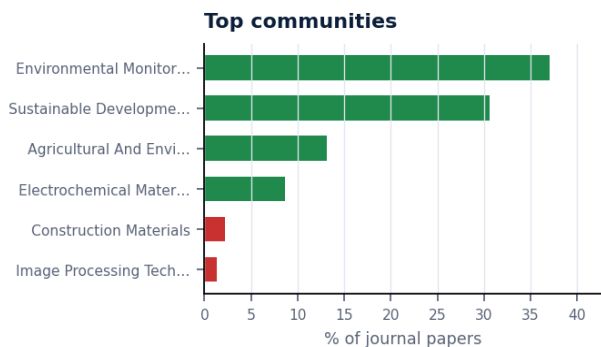
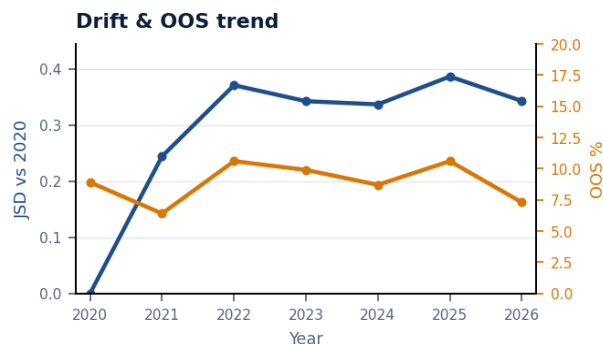
Right now, about 10% of Environmental Science's papers (678 out of 7,093) fall outside its core subject clusters — mostly work related to Construction Materials, Image Processing Techniques. This out-of-scope share has fallen since 2020. The spread of topics it covers has stayed fairly stable. Its largest single topic area today is “Environmental Monitoring.”

Scope split

90% in-scope

10% OOS

7,093 Papers	678 (9.6%) Out-of-scope	0.343 JSD 2026	-0.0436 Entropy Δ	3 Primary clusters	80.0% Primary coverage
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Community composition

Community	Status	Papers	Share of journal
Environmental Monitoring	In-scope	2,629	37.1%
Sustainable Development	In-scope	2,171	30.6%
Agricultural And Environmental Sciences	In-scope	926	13.1%
Electrochemical Materials	In-scope	608	8.6%
Construction Materials	Out-of-scope	156	2.2%
Image Processing Techniques	Out-of-scope	91	1.3%

Example papers

In-scope

- Niche expansion by indigenous fish species following the introduction of rainbow trout (*Oncorhynchus mykiss*) in a subtropical river system, upper Blyde River, South Africa
- Editorial: Microplastics in the Marine Environment: Sources, Distribution, Biological Effects and Socio-Economic Impacts
- Climate-driven microclimate dynamics and pollution-induced degradation forecasting in painted cave environmental systems
- Impact of Environmental Protection Regulations on Corporate Performance From Porter Hypothesis Perspective: A Study Based on Publicly Listed Manufacturing Firms Data
- Increasing objectivity associated with anadromous fish redd identification using a discriminant function analysis

Out-of-scope

- Selection of low impact development technical measures in the distribution area of expansive soil: a case study of Hefei, China
- Losses in fishery ecosystem services of the Dnipro river Delta and the Kakhovske reservoir area caused by military actions in Ukraine
- Effects of the annular eclipse on the surface O₃ in yunnan province, China
- Spatial Variation in Water Quality of the Burhi Gandak River: A Multi-Location Assessment
- Role of oil price volatility, energy efficiency, and financial stability on sustainable energy production

Frontiers in Robotics and AI

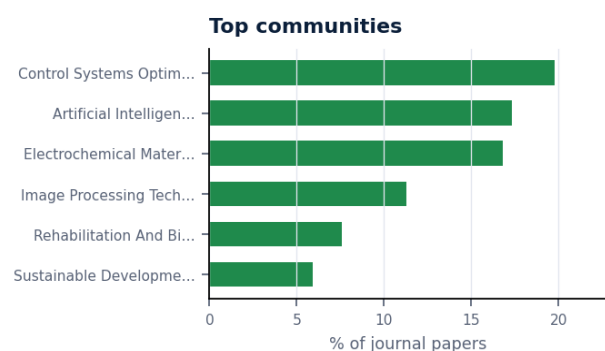
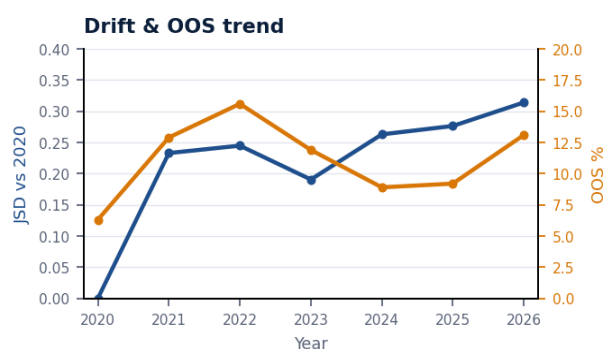
Moderate

Robotics and AI is showing a moderate amount of movement away from its 2020 starting point (JSD 0.31). This isn't dramatic, but the journal's content mix has evolved enough to be worth keeping an eye on.

Right now, about 12% of Robotics and AI's papers (207 out of 1,791) fall outside its core subject clusters — mostly work related to Space Weather Studies, Antenna Design. This out-of-scope share has risen since 2020. The spread of topics it covers has stayed fairly stable. Its largest single topic area today is "Control Systems Optimization."

Scope split

88% in-scope				12% OOS	
1,791 Papers	207 (11.6%) Out-of-scope	0.314 JSD 2026	+0.0379 Entropy Δ	7 Primary clusters	80.0% Primary coverage



Community composition

Community	Status	Papers	Share of journal
Control Systems Optimization	In-scope	354	19.8%
Artificial Intelligence Applications	In-scope	309	17.3%
Electrochemical Materials	In-scope	301	16.8%
Image Processing Techniques	In-scope	203	11.3%
Rehabilitation And Biomechanics	In-scope	136	7.6%
Sustainable Development	In-scope	106	5.9%

Example papers

In-scope

- Toward Accurate Visual Reasoning With Dual-Path Neural Module Networks
- Learning signs with NAO: humanoid robot as a tool for helping to learn Colombian Sign Language
- Toward an Attentive Robotic Architecture: Learning-Based Mutual Gaze Estimation in Human-Robot Interaction
- Translating Human Information into Robot Tasks: Action Sequence Recognition and Robot Control based on Human Motions
- A Virtual Sandbox Approach to Studying the Effect of Augmented Communication on Human-Robot Collaboration

Out-of-scope

- Automated capture and transfer of human facial expressions to humanoid robots for realistic patient simulation
- Editorial: Robotics, Autonomous Systems and AI for Nonurgent/Nonemergent Healthcare Delivery During and After the COVID-19 Pandemic
- Technological robot—Machine tool collaboration for agile production
- From the lab to the field with Evolutionary Field Robotics
- Workspace and trajectory-based experimental validation of a 3-armed 6-DOF parallel robot for femoral fracture surgery

Frontiers in Earth Science

Moderate

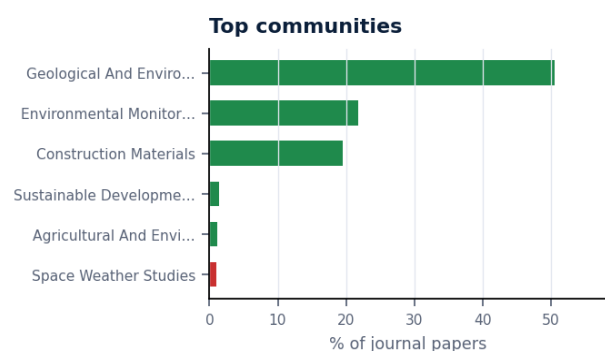
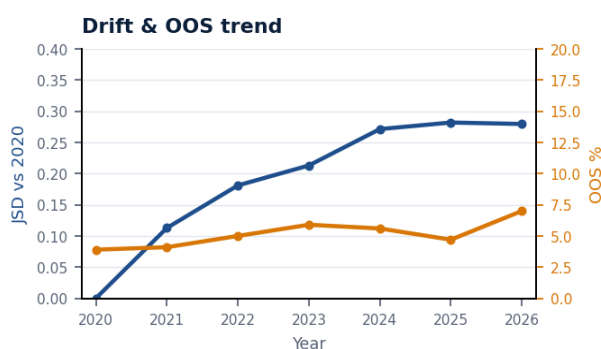
Earth Science is showing a moderate amount of movement away from its 2020 starting point (JSD 0.28). This isn't dramatic, but the journal's content mix has evolved enough to be worth keeping an eye on.

Right now, about 5% of Earth Science's papers (401 out of 7,870) fall outside its core subject clusters — mostly work related to Space Weather Studies, Image Processing Techniques. This out-of-scope share has risen since 2020. The journal is also spreading across a wider range of topics rather than staying tightly focused. Its largest single topic area today is “Geological And Environmental Sciences.”

Scope split

95% in-scope

7,870 Papers	401 (5.1%) Out-of-scope	0.280 JSD 2026	+0.1262 Entropy Δ	3 Primary clusters	80.0% Primary coverage
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Community composition

Community	Status	Papers	Share of journal
Geological And Environmental Sciences	In-scope	3,981	50.6%
Environmental Monitoring	In-scope	1,716	21.8%
Construction Materials	In-scope	1,534	19.5%
Sustainable Development	In-scope	109	1.4%
Agricultural And Environmental Sciences	In-scope	87	1.1%
Space Weather Studies	Out-of-scope	80	1.0%

Example papers

In-scope

- A Relative Paleointensity (RPI)-Calibrated Age Model for the Corinth Syn-rift Sequence at IODP Hole M0079A (Gulf of Corinth, Greece)
- The impact of Holocene deglaciation and glacial dynamics on the landscapes and geomorphology of Potter Peninsula, King George Island (Isla 25 Mayo), NW Antarctic Peninsula
- Factors contributing to morning rain in the upper Río Chagres Basin, Panamá
- Failure Characteristics and Treatment Measures of Tunnels in Expansive Rock Stratum
- How Does the Arctic Sea Ice Affect the Interannual Variability of Tropical Cyclone Activity Over the Western North Pacific?

Out-of-scope

- Intelligent Objective Osteon Segmentation Based on Deep Learning
- Automated Segmentation of Olivine Phenocrysts in a Volcanic Rock Thin Section Using a Fully Convolutional Neural Network
- Raw material base of Russia's unconventional oil and gas reserves (hydrocarbons shale strata)
- Red-green-bleached redox interfaces in the proximal Permian Cutler red beds: implications for regional fluid alteration
- The NEWTON-g Gravity Imager: Toward New Paradigms for Terrain Gravimetry

Frontiers in Chemistry

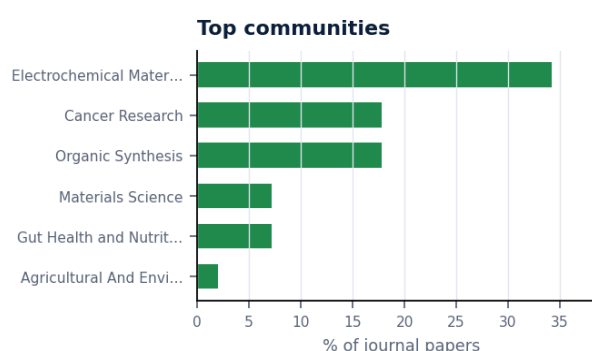
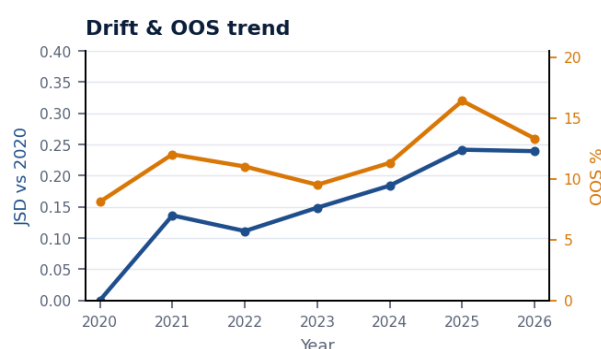
Low

Chemistry remains close to its original scope. Its content mix has changed only slightly since 2020 (JSD 0.24), which suggests the journal is publishing consistently within its intended focus.

Right now, about 11% of Chemistry's papers (649 out of 6,003) fall outside its core subject clusters — mostly work related to COVID-19 Health Impacts, Antimicrobial Resistance. This out-of-scope share has risen since 2020. The journal is also spreading across a wider range of topics rather than staying tightly focused. Its largest single topic area today is “Electrochemical Materials.”

Scope split

89% in-scope				11% OOS	
6,003 Papers	649 (10.8%) Out-of-scope	0.239 JSD 2026	+0.2844 Entropy Δ	5 Primary clusters	80.0% Primary coverage



Community composition

Community	Status	Papers	Share of journal
Electrochemical Materials	In-scope	2,051	34.2%
Cancer Research	In-scope	1,068	17.8%
Organic Synthesis	In-scope	1,067	17.8%
Materials Science	In-scope	434	7.2%
Gut Health and Nutrition	In-scope	430	7.2%
Agricultural And Environmental Sciences	In-scope	123	2.0%

Example papers

In-scope

- Mushroom tyrosinase enzyme catalysis: synthesis of larvicidal active geranylacetone derivatives against *Culex quinquesfasciatus* and molecular docking studies
- Photocatalytic removal of imidacloprid containing frequently applied insecticide in agriculture industry using Co₃O₄ modified MoO₃ composites
- Research Progress on the Photo-Driven Catalytic Production of Biodiesel
- Ferritin Conjugates With Multiple Clickable Amino Acids Encoded by C-Terminal Engineered Pyrrolysyl-tRNA Synthetase
- Layered metal sulfides with MaSbc- framework (M = Sb, In, Sn) as ion exchangers for the removal of Cs(■) and Sr(■) from radioactive effluents: a review

Out-of-scope

- Terazosin Analogs Targeting Pdgk1 as Neuroprotective Agents: Design, Synthesis, and Evaluation
- Deciphering the Impact of Mutations on the Binding Efficacy of SARS-CoV-2 Omicron and Delta Variants With Human ACE2 Receptor
- Metal interactions of α -synuclein probed by NMR amide-proton exchange
- Harnessing Brazilian biodiversity database: identification of flavonoids as potential inhibitors of SARS-CoV-2 main protease using computational approaches and all-atom molecular dynamics simulation
- Numerical study of the thermally stratified hemodynamic nanofluid flow with variable viscosity over a heated wedge

Frontiers in Surgery

Low

Surgery remains close to its original scope. Its content mix has changed only slightly since 2020 (JSD 0.23), which suggests the journal is publishing consistently within its intended focus.

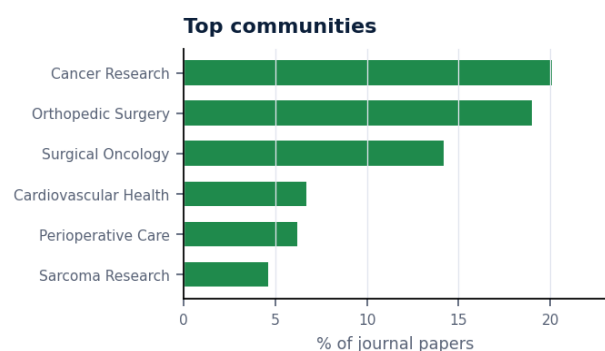
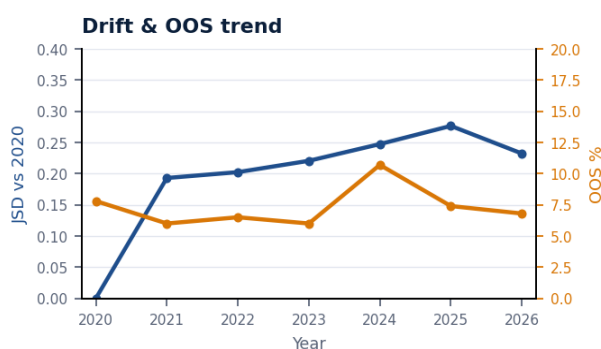
Right now, about 7% of Surgery's papers (407 out of 5,885) fall outside its core subject clusters — mostly work related to peripheral topics. This out-of-scope share has fallen since 2020. At the same time, the journal is concentrating more tightly around a smaller set of themes rather than spreading out. Its largest single topic area today is “Cancer Research.”

Scope split

93% in-scope

7% OOS

5,885 Papers	407 (6.9%) Out-of-scope	0.232 JSD 2026	-0.1047 Entropy Δ	10 Primary clusters	80.0% Primary coverage
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Community composition

Community	Status	Papers	Share of journal
Cancer Research	In-scope	1,181	20.1%
Orthopedic Surgery	In-scope	1,119	19.0%
Surgical Oncology	In-scope	834	14.2%
Cardiovascular Health	In-scope	397	6.7%
Perioperative Care	In-scope	362	6.2%
Sarcoma Research	In-scope	272	4.6%

Example papers

In-scope

- Analysis of 37 cases of percutaneous balloon compression for primary trigeminal neuralgia: experience and outcome from a single center
- Short-term outcomes of robotic vs. laparoscopic surgery for rectal cancer after neoadjuvant therapy: a meta-analysis
- Global research status and hotspot analysis of meniscal root tears based on the WOS database
- Comparison of robot versus fluoroscopy-assisted pedicle screw instrumentation in adolescent idiopathic scoliosis surgery: A retrospective study
- Risk Factors Analysis of Thoracic Trauma Complicated With Acute Respiratory Distress Syndrome and Observation of Curative Effect of Lung-Protective Ventilation

Out-of-scope

- Virtual Reality in the Neurosciences: Current Practice and Future Directions
- Fibular grafts in global reconstructive surgery: a bibliometric analysis
- Clinical application of ultrasound-guided acupotomy combined with platelet-rich plasma in the treatment of carpal tunnel syndrome
- Development and Application of Digital Maxillofacial Surgery System Based on Mixed Reality Technology
- Virtual-Augmented Reality and Life-Like Neurosurgical Simulator for Training: First Evaluation of a Hands-On Experience for Residents

Frontiers in Aging Neuroscience

Low

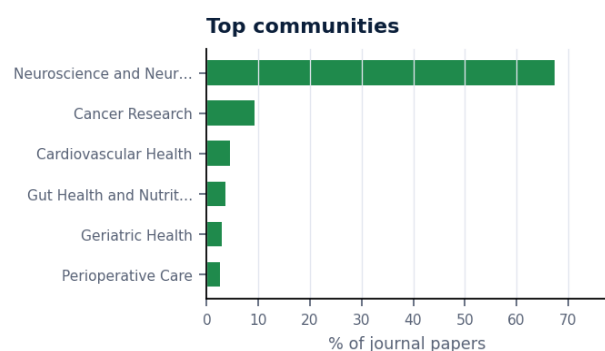
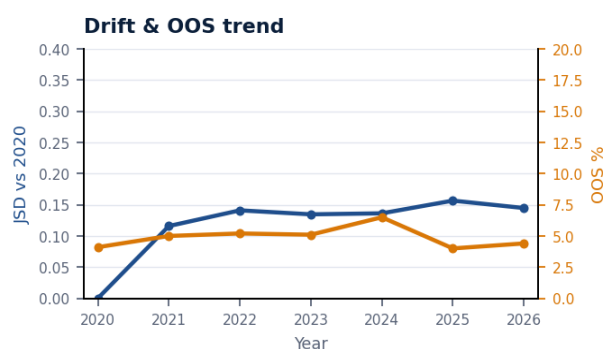
Aging Neuroscience remains close to its original scope. Its content mix has changed only slightly since 2020 (JSD 0.14), which suggests the journal is publishing consistently within its intended focus.

Right now, about 5% of Aging Neuroscience's papers (271 out of 5,420) fall outside its core subject clusters — mostly work related to Ocular Health, Neonatal And Surgical Interventions. This out-of-scope share has stayed similar since 2020. The journal is also spreading across a wider range of topics rather than staying tightly focused. Its largest single topic area today is "Neuroscience and Neurodegeneration."

Scope split

95% in-scope

5,420 Papers	271 (5.0%) Out-of-scope	0.145 JSD 2026	+0.1983 Entropy Δ	3 Primary clusters	80.0% Primary coverage
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Community composition

Community	Status	Papers	Share of journal
Neuroscience and Neurodegeneration	In-scope	3,652	67.4%
Cancer Research	In-scope	503	9.3%
Cardiovascular Health	In-scope	244	4.5%
Gut Health and Nutrition	In-scope	195	3.6%
Geriatric Health	In-scope	163	3.0%
Perioperative Care	In-scope	143	2.6%

Example papers

In-scope

- Dopamine 2 receptor ablation from cholinergic neurons attenuates L-DOPA induced dyskinesias
- Integrative analysis of plasma and substantia nigra in Parkinson's disease: unraveling biomarkers and insights from the lncRNA-miRNA-mRNA ceRNA network
- Effects of exercise-cognitive dual-task training on cognitive frailty in older adults: a randomized controlled trial
- Transcriptome Analysis of Microglia Reveals That the TLR2/IRF7 Signaling Axis Mediates Neuroinflammation After Subarachnoid Hemorrhage
- Older Age Increases the Amplitude of Muscle Stretch-Induced Cortical Beta-Band Suppression But Does not Affect Rebound Strength

Out-of-scope

- Homocysteine and Folic Acid: Risk Factors for Alzheimer's Disease—An Updated Meta-Analysis
- Glycyrrhizic Acid Improves Cognitive Levels of Aging Mice by Regulating T/B Cell Proliferation
- Risk Factors for Atorvastatin as a Monotherapy for Chronic Subdural Hematoma: A Retrospective Multifactor Analysis
- What happens in the brain when we die? Deciphering the neurophysiology of the final moments in life
- Hippocampal T1WI radiomics- and clinical feature-based models for predicting early mild cognitive impairment in secondary hydrocephalus